IPL Data Analysis – Assignment 2 Report

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1. Objective

The objective of this assignment was to clean, preprocess and analyze IPL match and ball-by-ball data. I also created useful features and visualizations to understand team, match, season and player performance.

2. Platform and Tools

I used **Google Colab and Python**. The main libraries used were **Pandas, NumPy, Matplotlib and Seaborn**.

3. Dataset Description

Two datasets were used:

- **matches.csv:** Contains match-level information such as teams, venue, toss winner, winner and result.
- **deliveries.csv:** Contains ball-by-ball information such as batsman, bowler, runs, wickets and overs.

Both datasets were connected using **match ID**.

4. Methodology

Load Data → Inspect Data → Clean Data → Standardize Team Names → Validate Data → Feature Engineering → Aggregation → Merge → EDA → Interpretation

5. Data Quality and Preprocessing

I checked missing values, duplicates, invalid values and unmatched match IDs. No duplicate rows or unmatched match IDs were found. I also checked negative run values and invalid wicket values.

Team names were standardized so that old and new franchise names were treated consistently. Outliers were examined but were not automatically removed because extreme values can represent genuine cricket matches.

6. Feature Engineering

New features were created, including:

- Season
- Toss advantage
- Match result type
- Match number in season
- Boundary indicator
- Dot-ball indicator
- Match phase: Powerplay, Middle and Death
- Run rate
- Boundary percentage
- Dot-ball percentage

Delivery data was aggregated by **match ID and batting team** and then merged with match-level data.

7. Exploratory Data Analysis

Different visualizations were created to study:

- Matches played per season
- Top venues
- Toss decisions and toss advantage
- Match result types
- Result margins
- First-innings scores
- Team winning percentages
- Runs, wickets and boundaries
- Powerplay, Middle and Death-over performance
- Correlations between numerical features
- Season-wise trends
- Top run scorers, wicket takers, four hitters and six hitters

8. Results and Discussion

Mumbai Indians had **144 wins**, Chennai Super Kings had **138 wins**, and Kolkata Knight Riders had **131 wins** in the processed data.

The toss winner also won the match in **554 matches**, while this did not happen in **541 matches**. This shows that winning the toss provided only a small overall advantage.

There were **578 wins by wickets** and **498 wins by runs**. The analysis also showed a relationship between boundaries and total runs. Phase-wise analysis showed that scoring patterns change between Powerplay, Middle and Death overs.

9. Limitations

The analysis has some limitations because team names and franchises changed over different IPL seasons. Different seasons also had different numbers of matches. Factors such as weather, pitch conditions, player injuries and team strategy were not available in the datasets.

10. Conclusion

In this assignment, IPL data was successfully cleaned, validated, feature-engineered and analyzed. The visualizations helped in understanding team performance, match results, scoring patterns, season trends and player performance. The processed dataset can also be used for further statistical or machine-learning analysis.

11. References

- CS3103 Machine Learning – Assignment 2
- IPL matches and deliveries datasets
- Pandas, NumPy, Matplotlib and Seaborn documentation

12. Code Link –

https://colab.research.google.com/drive/1tEiApaszOVFTMcFUbP_5lHm3sJpX0vZU#scrollTo=VExkj_Sam8nz

https://github.com/v-vaibhav07/ML/blob/main/LAB2/2401CS70_Lab2.ipynb